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Inventor(s)	Moore; Tamara et al.

Eight-dimensional barbed surgical thread

Abstract

An eight-dimensional barbed surgical thread including a central core and eight first barbs. The central core has a circumference, a proximal end and a distal end that is opposite the proximal end. The eight first barbs are positioned around the circumference of the central core so that the first barbs are radially adjacent to each other and obliquely extend from the central core.

Inventors: **Moore; Tamara (Hudson, WI), Wall; Laurie (Hudson, WI)**

Applicant: **Eurothreads LLC** (Hudson, WI)

Family ID: **74849304**

Assignee: **Eurothreads LLC (Cheyenne, WY)**

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Primary Examiner: Houston; Elizabeth

Assistant Examiner: Miller; Serenity A

Background/Summary

REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Applic. No. 62/898,174, which was filed on Sep. 10, 2019. The contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The invention relates generally to thread for surgical applications. More particularly, the invention relates to eight-dimensional barbed surgical thread.

BACKGROUND OF THE INVENTION

(2) A variety of aesthetic medical techniques have been developed to enhance the appearance of a person's skin and, in particular, the appearance of the person's face. These techniques range from cutting and tightening of the skin to injecting compositions into the skin such as to reduce wrinkles. Despite these advances, there is a continuing need to non-surgical techniques to improve a person's appearance.

SUMMARY OF THE INVENTION

(3) An embodiment of the invention is directed to an eight-dimensional barbed surgical thread that includes a central core and eight first barbs. The central core has a circumference, a proximal end and a distal end that is opposite the proximal end. The eight first barbs are positioned around the circumference of the central core so that the first barbs are radially adjacent to each other and obliquely extend from the central core.

(4) Another embodiment of the invention is directed to an eight-dimensional barbed surgical thread that includes a central core, eight first barbs and eight second barbs. The central core has a circumference, a proximal end and a distal end that is opposite the proximal end. The eight first barbs are positioned around the circumference of the central core and obliquely extend from the central core. Each of the eight first barbs has a distal end and a proximal end. The distal end of one of the first barbs is closer to the proximal end of the central core than the proximal end of the one of the second barbs. The eight second barbs are positioned around the circumference of the central core and obliquely extend from the central core. The eight second barbs are closer to the proximal end than the eight first barbs. Each of the eight second barbs has a distal end and a proximal end. The distal end of one of the second barbs is closer to the distal end of the central core than the proximal end of the one of the second barbs.

(5) Another embodiment of the invention is directed to a method of using an eight-dimensional barbed surgical thread. An eight-dimensional barbed surgical thread is provided that includes a central core, eight first barbs and eight second barbs. The central core has a circumference, a proximal end and a distal end that is opposite the proximal end. The eight first barbs are positioned around the circumference of the central core and obliquely extend from the central core. Each of the eight first barbs has a distal end and a proximal end. The distal end of one of the first barbs is closer to the proximal end of the central core than the proximal end of the one of the second barbs. The eight second barbs are positioned around the circumference of the central core and obliquely extend from the central core. The eight second barbs are closer to the proximal end than the eight first barbs. Each of the eight second barbs has a distal end and a proximal end. The distal end of one of the second barbs is closer to the distal end of the central core than the proximal end of the one of the second barbs. The eight-dimensional barbed surgical thread is inserted into a dermis on a patient. The eight first barbs engage the dermis to lift the dermis to a lifted position. The eight second barbs engage the dermis to anchor the dermis in the lifted position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of embodiments and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments and together with the description serve to explain principles of embodiments. Other embodiments and many of the intended advantages of embodiments will be readily appreciated as they become better understood by reference to the following detailed description. The elements of

the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding similar parts.
(2) FIG. **1** is a side view of an eight-dimensional barbed surgical thread according to an embodiment of the invention.
(3) FIG. **2** is a cross-sectional overview of the eight-dimensional barbed surgical thread.
(4) FIG. **3** is a cross-sectional view of the eight-dimensional barbed surgical thread illustrating locations of barbs extending therefrom.
(5) FIG. **4** is a front view before a procedure using the eight-dimensional barbed surgical thread.
(6) FIG. **5** is a front view after the procedure using the eight-dimensional barbed surgical thread.
(7) FIG. **6** is a side view before the procedure using the eight-dimensional barbed surgical thread.
(8) FIG. **7** is a side view after the procedure using the eight-dimensional barbed surgical thread.
(9) FIG. **8** is a front view after insertion of the eight-dimensional barbed surgical thread.
(10) FIG. **9** is front view after trimming of the eight-dimensional barbed surgical thread.
DETAILED DESCRIPTION OF THE INVENTION
(11) The invention is directed to an eight-dimensional barbed surgical thread as illustrated at **10** in FIG. **1**. The eight-dimensional barbed surgical thread **10** is particularly suited for use in cosmetic enhancement procedures. The eight-dimensional barbed surgical thread **10** includes a central core **20** from which a plurality of first barbs **22** extend.
(12) An advantage of the eight-dimensional barbed surgical thread **10** over the prior art barbed surgical threads is that the eight-dimensional barbed surgical thread **10** of this invention produces multiple effects once introduced into the dermis.
(13) These effects fall into three major categories: (1) instant skin lifting through mechanical effects, (2) cellular renewal through collagen stimulation and neovascularization to improve skin texture, fine lines and elasticity, and (3) skin tightening by contracting fat tissue.
(14) The eight-dimensional barbed surgical thread **10** is a medical device that in certain embodiments consists of a monofilament surgical suture. The eight-dimensional barbed surgical thread **10** may be fabricated from a variety of materials. In certain embodiments, the materials used to fabricate the eight-dimensional barbed surgical thread are absorbable inside of a human body. A non-limiting example of a suitable material for fabricating the eight-dimensional barbed surgical thread **10** is a polydioxanone (PDO)/polycaprolacton (PCL) polymer.
(15) In certain embodiments, a cut is used to form each of the first barbs **22**. A variety of cutting techniques may be used. In certain embodiments, the eight-dimensional barbed surgical thread **10** is fabricated by laser cutting into the surgical suture octagonally at an angle of about 45 degree as illustrated in FIG. **2**. The locations of the eight cuts is indicated by letters A-H.
(16) Using such a configuration, the eight first barbs **22** are radially adjacent to each other when going around a circumference of the eight-dimensional barbed surgical thread **10**. While in certain embodiments, the first barbs **22** are immediately radially adjacent to each other such as illustrated in FIG. **2**, it is possible that there can be a spacing between adjacent first barbs **22**. As used herein, the spacing between adjacent first barbs **22** is less than about 20 percent of the width of the widest portion of the first barbs **22**. In other embodiments, the spacing between adjacent first barbs **22** is less than about 10 percent of the width of the widest portion of the first barbs **22**.
(17) Using the preceding configuration enhances the potential of contacts between the first barbs **22** after insertion into the dermis regardless of the orientation of the eight-dimensional barbed surgical thread **10**, which reduces the potential of the dermis sagging after insertion of the eight-dimensional barbed surgical thread **10**.
(18) The cutting that is done to form the first barb **22** is done at an angle to the surface of the central core **20**. In certain configurations, the first barbs **22** are oblique, notched protrusions formed into a surgical suture at an angle of between about 10 degrees and about 15 degrees. In other embodiments, the angle of the cut to form the first barb **22** is about 12.5 degrees. Cutting in this manner causes a width of the first barb **22** decreases when moving from the proximal end **40*a*** to

the distal end **42***a*. In certain embodiments, the distal end **42***a* of the first barb **22** is pointed.
(19) In certain embodiments, a ratio of a length of the barb to a width of the barb is between about 2:1 and about 5:1. Forming the first barbs **22** with the length to width ratio in this range provides the first barbs **22** with sufficient strength for the first barbs **22** to perform the desired functions during the insertion process and after insertion.
(20) Adjacent first barbs **22** may be offset in a direction moving from a distal end **32** to a proximal end **30** of the central core **20** as illustrated in FIG. **1**. In one such embodiment, a distal end **42***a* of one first barb **22***a* is approximately aligned with a proximal end **40***a* of an adjacent first barb **22***b* around a circumference of the central core **20**, as illustrated in FIG. **1**. Additionally, the distal end **42***a* of every other first barb **22***a*, **22***c* is approximately aligned around a circumference of the central core **20**, as illustrated in FIG. **1**.
(21) The first barbs **22** have a proximal end **40***a* and a distal end **42***a*. The proximal end **40***a* is where the first barb **22** attaches to the central core **20**. The distal end **42***a* is opposite the proximal end **40**.
(22) The plurality of first barbs **22** are cut at intervals of about 45 degrees on eight sides of the eight-dimensional barbed surgical thread **10**. In certain embodiments, the distance between adjacent first barbs **22** is about 1.5 millimeters, which is closer than the conventional spacing of about 1.8 millimeters that is used in the prior art surgical thread that is used when performing facelifts.
(23) The plurality of first barbs **22** are oriented such that the distal end **42***a* of each first barb **22** is closer to the central core distal end **32** than the proximal end **40***a* of each first barb **22**. As described in more detail below, a primary function of the plurality of first barbs **22** is lifting.
(24) In certain embodiments, the plurality of first barbs **22** occupy greater than about ½ of a length of the eight-dimensional barbed surgical thread **10**. In other embodiments, the plurality of first barbs **22** occupy about ⅔ of the length of the eight-dimensional barbed surgical thread **10**.
(25) The eight-dimensional barbed surgical thread **10** is typically inserted so that the plurality of first barbs **22** are facing upwardly. The configuration of the plurality of first barbs **22** on the eight-dimensional barbed surgical thread **10** thereby allows for most of the eight-dimensional barbed surgical thread **10** to assist in lifting of the tissue.
(26) The eight-dimensional barbed surgical thread **10** also includes a plurality of second barbs **24** proximate the central core distal end **32**. The eight-dimensional barbed surgical thread **10** is typically inserted so that the plurality of second barbs **24** are facing downwards. As described in more detail below, a primary function of the plurality of second barbs **24** is anchoring.
(27) The plurality of second barbs **24** are cut at intervals of about 45 degrees on eight sides of the eight-dimensional barbed surgical thread **10**. In certain embodiments, the distance between adjacent second barbs **24** is about 1.5 millimeters, which is closer than the conventional spacing of about 1.8 millimeters that is used in the prior art surgical thread that is used when performing facelifts.
(28) Other than the orientation, the second barbs **24** may be shaped similarly to the first barbs **22**. The plurality of second barbs **24** are oriented such that the distal end **42***b* of each second barb **24** is closer to the distal central core proximal end **30** than the distal proximal end **40***b* of each second barb **24**.
(29) In certain embodiments, the plurality of second barbs **24** occupies less than about ½ of the length of the eight-dimensional barbed surgical thread **10**. In other embodiments, the plurality of second barbs **24** occupies about ⅓ of the length of the eight-dimensional barbed surgical thread **10**.
(30) In certain embodiments, there may be an intermediate region **34** of the eight-dimensional barbed surgical thread **10** that is intermediate the upwardly facing first barbs **22** and the downwardly facing second barbs **24** from which no barbs extend therefrom. This intermediate region **34** on the eight-dimensional barbed surgical thread **10** may have a length that is smaller than the length of the eight-dimensional barbed surgical thread **10** over which the upwardly extending

first barbs **22** extend. The length of the intermediate region of the eight-dimensional barbed surgical thread **10** may be smaller than the length of the eight-dimensional barbed surgical thread **10** over which the downwardly extending second barbs **24** extend.
(31) Unlike the prior art two-dimensional, three-dimensional and four-dimensional barbed surgical thread that limited the ability to engage tissue, the eight-dimensional barbed surgical thread **10** of this invention significantly improves the viability and uniformity of tissue engagement thus the sustainability of the desired result.
(32) Additionally, upon insertion of the eight-dimensional barbed surgical thread **10** into the dermis, the dermis sustains minor injuries. These minor injuries engage the body's natural healing process and stimulate the skin cells to produce collagen and blood vessels, which improves skin microcirculation.
(33) It has been found that in areas of the body that do not have bone structure beneath, such as the fatty area of the cheek, having more points of tissue attachment provides a greater degree of lift, greater points of injury and a larger amount of fiberblasting, which thereby results in greater collagen production.
(34) The eight-dimensional barbed surgical **10** is inserted into the subcutaneous layer of the dermis. In certain embodiments, a cannula is used to insert the eight-dimensional barbed surgical thread **10** into the dermis. In other embodiments, the cannula is a rounded or L-tipped surgical steel cannula.
(35) The eight-dimensional barbed surgical thread **10** is not attached to the surgical cannula and is introduced into the soft tissue through the tip of the cannula. Once the cannula reaches the desired end site, the cannula is rotated about 180 degrees mechanically engaging the eight-dimensional barbed surgical thread **10** at the distal (lower) end.
(36) The cannula is then removed leaving only the eight-dimensional barbed surgical thread **10** in the dermis. Approximating the tissue up the eight-dimensional barbed surgical thread **10** (lifting the tissue onto each barb) allows for each of the first barbs **22** to become securely engaged in the dermis. Once the desired result is achieved, the eight-dimensional barbed surgical thread **10** is anchored at the proximal (upper) end. Any excess portion of the eight-dimensional barbed surgical thread **10** is cut off and discarded.
(37) The use of eight-dimensional barbed surgical threads **10** is indicated for soft tissue augmentation where the insertion of surgical sutures is appropriate. The eight-dimensional barbed surgical threads **10** are used to lift, contour and volumize the skin. The implantation of the eight-dimensional barbed surgical threads **10** is indicated for subcutaneous (intradermal and hypodermal) implantation.
(38) In operation, once the area to be treated is defined, and an appropriate examination is completed, the patient is seated. The treatment area should be prepped by cleansing and removing the topical anesthetic. If topical anesthetic is to be used, it is applied liberally to the treatment areas.
(39) The appropriate thread packages required for the treatment area are opened and removed from the package. The thread is attached in accordance with the manufacturer's instructions. Proper use of the product should minimize the chances of dislodging or breaking while injecting the thread.
(40) Correct injection technique is important to the success of the treatment in achieving the desired results. The needle or cannula should be inserted into the treatment site with the tip ending up at an appropriate depth within the skin. The eight-dimensional barbed surgical thread **10** should then be released using a slow, steady withdrawal of the needle/cannula. Overcorrection, which is more threads than suggested or required, is generally not needed and is to be avoided.
(41) Once the first thread is appropriately inserted, another thread is inserted into the next adjacent location, and the process is repeated. Care should be taken to adequately assess the entire area to be treated with the correct number of threads to ensure even and symmetrical distribution of the product.
(42) Once the injection is completed, the treated areas should be gently massaged, setting the threads per instruction. More vigorous massage may result in additional swelling, bruising or

dislodgement of the thread.
(43) Unlike the prior art two-dimensional, three-dimensional and four-dimensional barbed surgical threads that are limited in their ability to engage tissue due to barbs being cut respectively on only 2, 3 or 4 sides of the thread. This greatly limits the number of points of engagement into soft tissue. It is this engagement that is necessary to achieve a sustainable lift.
(44) With eight-dimensional barbed surgical threads **10**, the presence of additional first barbs **22** more than double the points of tissue engagement of the four-dimensional barbed surgical thread. Regardless of how the eight-dimensional barbed surgical thread **10** is inserted into subcutaneous tissue, the additional first barbs **22** allows the eight-dimensional barbed surgical thread **10** to grab onto additional dermis resulting in a better lift.
(45) The eight-dimensional barbed surgical thread that is fabricated from PDO alone or in combination with PCL is slowly absorbed after implantation into the patient. In certain embodiments, the eight-dimensional barbed surgical thread is resorbed over a time period of between about 180 days and about 240 days. After this time period, the eight-dimensional barbed surgical thread **10** is substantially absorbed into the surrounding tissue. As used herein, substantially absorbed means that there are only minimal traces remaining of the eight-dimensional barbed surgical thread **10**.
Examples
(46) In the following example, the performance of the eight-dimensional barbed surgical thread according to this invention is evaluated. The subject on which the procedure was performed is a 60+-year-old Caucasian man. The subject complained of sagging skin such as on his chest.
(47) The concerns of the male subject are similar to the concerns of many 60 to 70 year old men.
(48) FIGS. **4** and **5** are before and after front photographs, respectively, of a male subject on which mastopexy was performed while FIGS. **6** and **7** are before and after side photographs, respectively of the male subject on which the mastopexy was performed.
(49) FIG. **8** illustrates proximal ends of the eight-dimensional barbed surgical threads **10** extending from the patient **100**. After the proximal ends of the eight-dimensional barbed surgical threads **10** are trimmed, the final lifted results are illustrated in FIG. **9**.
(50) Cosmetic medical procedures were conducted to evaluate the performance of the eight-dimensional barbed surgical thread of this application to the prior art four-dimensional barbed surgical thread as well as to a six-dimensional barbed surgical thread that is the subject of a separate patent application that is being filed on the same day as this patent application.
(51) The performance of the respective surgical threads as evaluated with respect to lifting of the dermis and stimulating collagen production. The relative performance of the six-dimensional barbed surgical thread and the eight-dimensional barbed surgical thread with respect to the prior art four-dimensional barbed surgical thread was classified as either no change, minimal change, moderate change and significant change from the perspective of an experienced medical practitioner in the field of cosmetic medical procedures.
(52) The results from the use of the six-dimensional barbed surgical thread, the eight-dimensional barbed surgical thread and the prior art four-dimensional barbed surgical thread in conjunction with fifteen different cosmetic medical procedures are classified as:
(53) (1) Six-dimensional barbed surgical thread and eight-dimensional barbed surgical thread exhibit similar performance to each other and superior performance when compared to the prior art four-dimensional barbed surgical thread (Table 1)
(54) (2) Six-dimensional barbed surgical thread exhibits superior performance to the eight-dimensional barbed surgical thread and both the six-dimensional barbed surgical thread and the eight-dimensional barbed surgical thread exhibit superior performance to the prior art four-dimensional barbed surgical thread (Table 2).
(55) (3) Eight-dimensional barbed surgical thread exhibits superior performance to the six-dimensional barbed surgical thread and both the six-dimensional barbed surgical thread and the

eight-dimensional barbed surgical thread exhibit superior performance to the prior art four-dimensional barbed surgical thread (Table 3).
(56) (4) Six-dimensional barbed surgical thread, eight-dimensional barbed surgical thread and the prior art four-dimensional barbed surgical thread exhibit similar performance to each other (Table 4).
(57) TABLE-US-00001 TABLE 1 Similar Superior Performance of 6D and 8B Barbed Thread Treatment type Description 4D vs 6D Barb 4D vs 8D Barb Mid-Rhytidectomy Midface to temple lift significant significant Abdominplasty Abdominal lift moderate moderate Brachioplasty Upper arm lift moderate moderate Kneeplasty/Thighplasty Knee/lower thigh lift significant significant
(58) TABLE-US-00002 TABLE 2 Superior Performance of 6D Barbed Thread Treatment type Description 4D vs 6D Barb 4D vs 8D Barb Nasolabial Folds Removal Smile Lines significant moderate Platysmaplasty Neck Skin Tightening significant moderate
(59) TABLE-US-00003 TABLE 3 Superior Performance of 8D Barbed Thread Treatment type Description 4D vs 6D Barb 4D vs 8D Barb Lower Rhytidectomy Jawline and neck lift moderate significant Gyneocomastia Male breast size reduction minimal moderate Mastopexy Size > B+ Breast lift moderate significant Mastopexy Size > C+ Breast lift minimal moderate Gluteal Lift Buttock lift minimal moderate Jawline Contouring Jowl lift moderate significant
(60) TABLE-US-00004 TABLE 4 Similar Performance of 6D, 8D and 4D Barbed Thread Treatment type Description 4D vs 6D Barb 4D vs 8D Barb Browplasty Brow or forehead lift no change no change Blepharoplasty Eye hood lift no change no change Mastopexy Size > D Breast lift no change minimal
(61) In view of the fact that the six-dimensional barbed surgical thread and the eight-dimensional barbed surgical thread do not exhibit superior results to the prior art four-dimensional barbed surgical thread for all of the cosmetic medical procedures set forth in Tables 1-4 and in view of the fact that the six-dimensional barbed surgical thread and the eight-dimensional barbed surgical thread exhibit similar performance in certain cosmetic medical procedures but the six-dimensional barbed surgical thread provides superior results in some cosmetic medical procedures and inferior results in some cosmetic medical procedures as compared to the eight-dimensional barbed surgical thread indicate that the eight-dimensional barbed surgical thread is a not obvious modification of the prior art four-dimensional barbed surgical thread as well as the six-dimensional barbed surgical thread.
(62) In the preceding detailed description, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as “top,” “bottom,” “front,” “back,” “leading,” “trailing,” etc., is used with reference to the orientation of the Figure(s) being described. Because components of embodiments can be positioned in a number of different orientations, the directional terminology is used for purposes of illustration and is in no way limiting. It is to be understood that other embodiments may be utilized and structural or logical changes may be made without departing from the scope of the present invention. The preceding detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present invention is defined by the appended claims.
(63) It is contemplated that features disclosed in this application, as well as those described in the above applications incorporated by reference, can be mixed and matched to suit particular circumstances. Various other modifications and changes will be apparent to those of ordinary skill.

Claims

1. A method of using an eight-dimensional barbed surgical thread to perform one of a lower rhytidectomy, a gyneocomastia, a mastopexy with a breast size less than B+, a mastopexy with a breast size of less than C+, a gluteal lift, and jawline contouring, wherein the method comprises:

providing the eight-dimensional barbed surgical thread comprising a central core, eight first barbs and eight second barbs, wherein the central core has a circumference, a central core proximal end and a central core distal end that is opposite the central core proximal end, wherein the eight first barbs are positioned radially adjacent to each other around the circumference of the central core and obliquely extend from the central core, wherein each of the eight first barbs has a first barb distal end and a first barb proximal end, wherein each of the eight first barbs is oriented so that the first barb distal end is closer to the central core distal end than the first barb proximal end, wherein the eight second barbs are positioned radially adjacent to each other around the circumference of the central core and obliquely extend from the central core, wherein the eight second barbs are closer to the central core distal end than the eight first barbs, wherein each of the eight second barbs has a second barb distal end and a second barb proximal end and wherein each of the second barbs is oriented so that the second barb distal end is closer to the central core proximal end than the second barb proximal end; and inserting the eight-dimensional barbed surgical thread into a dermis on a patient; engaging the eight first barbs in the dermis to lift the dermis to a lifted position; and engaging the eight second barbs in the dermis to anchor the dermis in the lifted position and complete the one of the lower rhytidectomy, the gyneocomastia, the mastopexy with the breast size less than B+, the mastopexy with the breast size of less than C+, the gluteal lift, and the jawline contouring.
2. The method of claim 1, wherein the eight first barbs occupy about ⅔ of a length of the eight-dimensional barbed surgical thread and wherein the eight second barbs occupy about ⅓ of the length of the eight-dimensional barbed surgical thread.
3. The method of claim 1, wherein adjacent barbs in the eight first barbs are offset in a direction extending between the proximal end and the distal end of the central core so that the first barb distal end of one of the eight first barbs is approximately aligned with the first barb proximal end of an adjacent one of the eight first barbs around a circumference of the central core and wherein every other of the eight first barbs is approximately aligned around a circumference of the central core.
4. The method of claim 1, wherein each of the eight first barbs occupies about 45 degrees of a circumference of the central core and wherein a cut is formed at an angle of between about 10 degrees and about 15 degrees with respect to a surface of the central core to form each of the eight first barbs.
5. The method of claim 1, wherein the eight-dimensional barbed surgical thread comprises a plurality of rows of the eight first barbs and wherein a spacing between the rows of the eight first barbs is less than about a length of one of the eight first barbs in a direction that extends between the central core proximal end and the central core distal end.
6. The method of claim 1, wherein the eight-dimensional barbed surgical thread is inserted into the dermis using a cannula.
7. The method of claim 6, wherein the cannula is rotated to cause the eight second barbs to engage the dermis.
8. The method of claim 1, wherein insertion of the eight-dimensional barbed surgical thread causes cellular renewal through collagen stimulation and neovascularization to improve skin texture.
9. The method of claim 1, wherein insertion of the eight-dimensional barbed surgical thread into the dermis causes skin tightening caused by contracting fat tissue.
10. The method of claim 1, wherein after anchoring the dermis in the lifted position, cutting off a portion of the proximal end of the eight-dimensional barbed surgical thread that extends through a skin surface on the person.
11. A method of performing one of a lower rhytidectomy, a gyneocomastia, a mastopexy with a breast size less than B+, a mastopexy with a breast size of less than C+, a gluteal lift, and jawline contouring, wherein the method comprises: providing an eight-dimensional barbed surgical thread comprising a central core, eight first barbs and eight second barbs, wherein the central core has a

circumference, a central core proximal end and a central core distal end that is opposite the central core proximal end, wherein the eight first barbs are positioned radially adjacent to each other around the circumference of the central core and obliquely extend from the central core, wherein each of the eight first barbs has a first barb distal end and a first barb proximal end, wherein each of first barbs is oriented so that the first barb distal end is closer to the central core distal end than the first barb proximal end, wherein the eight second barbs are positioned radially adjacent to each other around the circumference of the central core and obliquely extend from the central core, wherein the eight second barbs are closer to the central core distal end than the eight first barbs, wherein each of the eight second barbs has a second barb distal end and a second barb proximal end and wherein each of the second barbs is oriented so that the second barb distal end is closer to the central core proximal end than the second barb proximal end; providing a cannula; positioning the eight-dimensional surgical thread at least partially in the cannula; inserting the cannula into the dermis; remove the cannula from the dermis while leaving the eight-dimensional surgical thread at least partially in the dermis; engaging the eight first barbs in the dermis to lift the dermis to a lifted position; and engaging the eight second barbs in the dermis to anchor the dermis in the lifted position and complete the one of the lower rhytidectomy, the gyneocomastia, the mastopexy with the breast size less than B+, the mastopexy with the breast size of less than C+, the gluteal lift, and the jawline contouring.
12. The method of claim 11, wherein the cannula is rotated to cause the eight second barbs to engage the dermis.
13. The method of claim 11, wherein the cannula is a rounded or L-tipped surgical steel cannula.
14. The method of claim 11, wherein after anchoring the dermis in the lifted position, cutting off a portion of the proximal end of the eight-dimensional barbed surgical thread that extends through a surface of the skin.